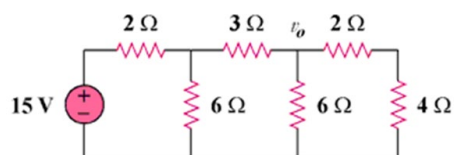


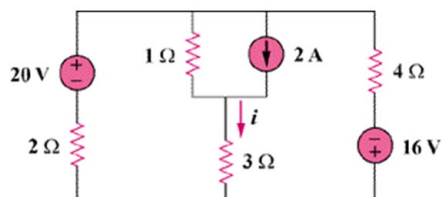
ENGR 2303: Electrical Science

HW 4_Ch4

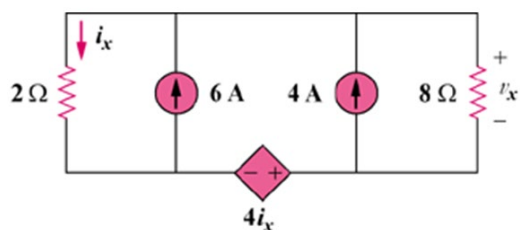
- 1) For the circuit below, assume $v_o = 1$ V, and use linearity to find the actual value of v_o .



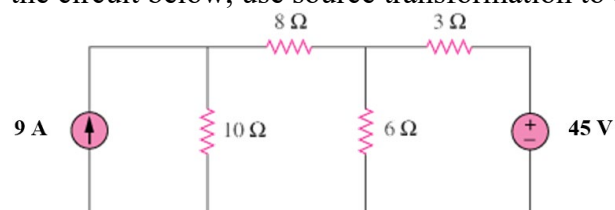
- 2) For the circuit below, use superposition to find i . Calculate the power delivered to the 3-Ω resistor.



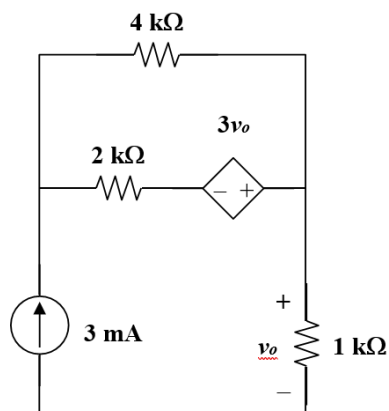
- 3) Use superposition to solve for v_x in the circuit below.



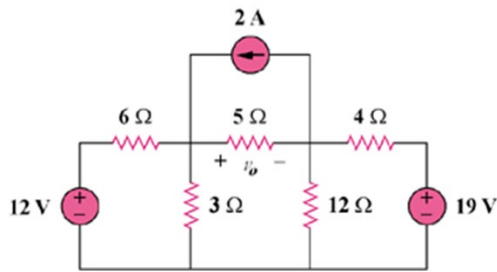
- 4) For the circuit below, use source transformation to determine the current and power in the 8-Ω resistor.



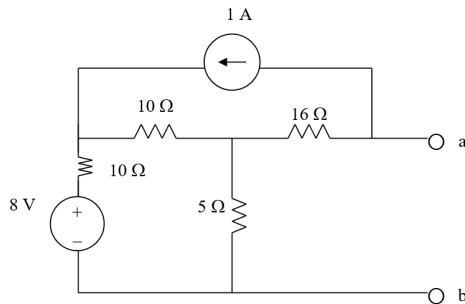
- 5) Use source transformation to find v_o in the circuit below.



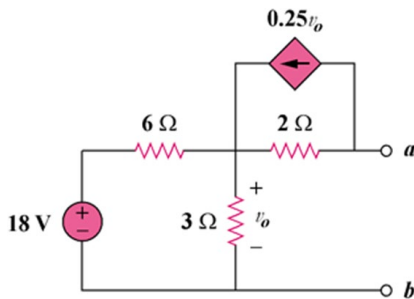
- 6) Use Thevenin's theorem and superposition principle to find v_o in the circuit below



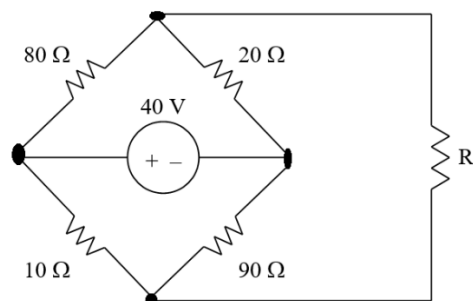
- 7) Obtain the Thevenin equivalent at terminals a-b of the circuit below.



- 8) Find the Norton equivalent at terminals **a-b** of the circuit below.



- 9) The variable resistor R in the circuit below is adjusted until it absorbs the maximum power from the circuit. (a) Calculate the value of R for maximum power. (b) Determine the maximum power absorbed by R .



- 10) A transducer is modeled with a current source I_s and a parallel resistance R_s . The current at the terminals of the source is measured to be 9.97506 mA when an ammeter with an internal resistance of $20\ \Omega$ is used.
- If adding a $2\text{-k}\Omega$ resistor across the source terminals causes the ammeter reading to fall to 9.87654 mA, calculate I_s and R_s .
 - What will the ammeter reading be if the resistance between the source terminals is changed to $4\text{ k}\Omega$?